Detecting AI-Generated Vietnamese Text

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Abstract

Texts created by AI gain popularity, causing individuals to struggle with writing, making it impossible to identify, and negatively influencing public perception. Some of the content was generated by AI without emotions, resulting in contributions on social media that lack depth and induce negative or incorrect awareness for users. To address this issue, we developed a model for distinguishing texts in order to improve the quality of replies on social media and the web, allowing users to contribute to the creation of comprehensive knowledge about topics such as Wikipedia. We conducted evaluations using state-of-the-art methods, including ViT5, BartPho, PhoBERT, mDeberta V3, and mBERT. These results contribute to detect texts generated by AI on social media. Furthermore, research lays the foundation for future advancements in AI-generated text detection and provides valuable insights for researchers in the field of natural language

Keywords: AI-generated text detection, Vietnamese language processing, large language models, dataset construction, machine learning

1. Introduction

The rapid growth of Large Language Models (LLMs) has significantly improved the quality of AI-Generated Texts (AIGTs), allowing models such as GPT, Gemini, and other chatboxes to be used in everyday life to produce high-quality texts such as academic writing, question-answering, and translation. These AIGTs are often indistinguishable from Human-Written Texts (HWTs), making AIGT detection an important but difficult challenge for effective categorization. Moreover, AIGTs might be intentionally employed for manipulating information or spreading false news, which could lead to significant societal consequences. To gain a clearer understanding of how widespread AIGTs are on social media platforms, we seek to measure and track their occurrence, posing the question: Are we currently engaging with AI-generated texts on social media?

Vietnamese text data from newspapers is usually used in the data; AI is then used for data refining and tagging. In this work, we employ prompts for a range of major language models from 2022 to the present, as well as the user-posed query on Noron and the headings of various topics on Wikipedia, to efficiently recognize texts. Additionally, we divide the amount of data produced by AI unevenly to guarantee that bias is eliminated, otherwise the model will readily pick up features from the data and improve its accuracy.

To quantify and monitor AIGTs on social media, we collect textual data from 2 popular platforms spanning as most LLMs are released after 2022. After data preprocessing, we obtain over 50,000 posts from Noron and 4,000 contents from Wikipedia. After that, we filter the relevant topics to get rid of any sensitive information and make a prompt for AI to produce or improve. We name it as **VSM-D**, short for Vietnamese Social Media Dataset.

Based on VSM-D, we use a language model for Vietnamese consisting of PhoBert, ViT5, and Cafe Bert to create a tool that can detect AI on social media, with accuracy peaking at 90% and a website to demo the results.

2. Related Work

A comprehensive understanding of high-quality datasets and text detection methodologies is essential for enhancing the effectiveness of AI-generated text detection research. This section reviews prominent global and domestic research efforts, thereby identifying the research gaps addressed by this study.

2.1. Global Landscape

2.1.1 Former LLM-Generated Text Detection Datasets

One of the earliest initiatives is the **Human ChatGPT Comparison Corpus (HC3)** [1], designed to compare ChatGPT responses with human writing in both English and Chinese. However, HC3 faces limitations regarding the diversity of generated instructions. Other specialized datasets include **CHEAT** [2], which focuses on detecting fraudulent academic abstracts, and **TweepFake** [3], devel-

oped to identify deepfake tweets on social media platforms like Twitter.

2.1.2 LLM-Generated Text Detection

- **DetectGPT**: [4] Mitchell et al. proposed **DetectGPT**, a zero-shot method that analyzes the curvature of a model's log-probability function to identify machine-generated text without additional training.
- **Watermark Technology:** Watermarking techniques are an indispensable component in detecting AI-generated images, contributing significantly to the protection of intellectual property and ownership in the visual arts. With the emergence of Large Language Models (LLMs), these techniques have been adapted to identify machine-generated text. Current methodologies applying watermarking include statistical methods [6] and secret key-based watermarking technologies [7].
- **Fine-tuning LMs Methods:** This strategy involves the supervised adaptation of Transformer-based models to effectively differentiate between machine-generated and human-written text. By leveraging the deep Natural Language Understanding (NLU) of pre-trained architectures, researchers append a specialized classification layer to the model to perform high-precision text categorization. This methodology requires paired datasets for training and has been widely proven by the research community to deliver superior performance in identifying subtle linguistic nuances. [8]
- **ChatGPT Detector:**[1] Proposed by Guo et al., this detector is specifically designed to differentiate between human-written content and ChatGPT-generated responses. The model is built by fine-tuning a RoBERTa architecture using the HC3 dataset. The researchers provided two distinct training approaches for the RoBERTa model: the first utilizes only the pure generated text, while the second leverages joint training with question-answer pairs. To maintain methodological consistency with other detection systems, evaluations often focus on the pure text-based approach.
- **LMDetector**: Another significant model-based approach is the development of detectors through the fine-tuning of pre-trained language models (LMs) augmented with an extra classification head. A typical implementation of this method utilizes models such as BERT to assess detection performance, following the experimental framework established by Ippolito et al. [9]
- **MGTBench:** [10] This is the first comprehensive benchmark framework designed for machine-generated text (MGT) detection against powerful Large Language Models (LLMs) such as ChatGPT-turbo and Claude. MGTBench employs a modular architecture comprising an input module, a detection module, and an evaluation module. The framework supports 13 distinct detection methodologies, which are categorized into eight metric-based approaches and five model-based approaches. Furthermore, it facilitates the analysis of detection robustness by simulating various adversarial attacks, such as paraphrasing and adversarial perturbations.

2.2. Vietnamese Landscape

Research on AI-generated text detection in Vietnam remains relatively nascent. A pioneering study in this field is the **ViDetect** [11] dataset, which comprises approximately 6,800 Vietnamese essay samples. This dataset consists of human-written essays collected before 2021 and AI-generated versions rewritten by models such as ChatGPT 3.5 and 4.0. The authors evaluated several state-of-the-art Vietnamese language models, including ViT5, BartPho, PhoBERT, mDeBERTa V3, and mBERT. Results indicate that these models, particularly when combined with fine-tuning techniques, demonstrate strong classification performance.

AI News Detection with Multilingual-E5 and BERT: [12] A recent study by Huynh et al. has expanded the detection scope to the field of journalism and news. This research curated a massive dataset of 200,000 Vietnamese news articles, combining reputable sources like Thanh Niên and VnExpress with content generated by advanced LLMs such as GPT-4o Mini and Gemini Flash 1.5. By utilizing a hybrid architecture of Multilingual-E5 embeddings and Transformer models (including RoBERTa and DeBERTaV3), the study achieved a superior accuracy exceeding 99% in identifying AI-generated news

Although studies like ViDetect and the news detection research by Huynh et al. have achieved impressive results, they primarily focus on texts with a formal tone and rigid structures, such as essays and journalistic articles. A sig-

nificant gap remains when applying these models to the social media domain, where language is often informal, featuring abbreviations, slang, and high emotional expressiveness. Current research acknowledges that transitioning to less formal text types or online conversations still requires substantial adaptation in both data and modeling While the latest research has begun integrating modern LLMs like GPT-4o Mini and Gemini Flash 1.5, the risk of models becoming obsolete due to rapid technological evolution (version drift) persists. Furthermore, detecting sophisticatedly edited AI content or paraphrased material derived from human writing on social platforms remains a significant challenge.

Consequently, our study introduces the VSM-D, focusing on content from Noron and Wikipedia. This research goes beyond simple text classification by aiming to improve interaction quality and community knowledge. It seeks to identify AI-generated content that lacks depth or emotional resonance in cyberspace.

3. Background

3.1. Vietnamese Linguistic Characteristics

Vietnamese, a member of the Austroasiatic family, exhibits fundamental differences from inflectional European languages. We focus on three key aspects that directly influence model design and preprocessing strategies:

- **Isolating Morphology and Word Segmentation:** Unlike inflectional languages where words change form to indicate grammar (e.g., adding ”s” or ”ed”), Vietnamese words are invariant. The fundamental unit is the monosyllable; consequently, meaningful lexical units are often compounds of multiple syllables. Crucially, whitespace serves as a syllable delimiter rather than a word boundary. This renders word segmentation a mandatory and non-trivial preprocessing step to identify semantic units correctly, distinguishing it from whitespace-delimited languages.

- **Phonology and Orthography:** Vietnamese is defined by a rigorous tonal system (typically six tones), where pitch contour dictates lexical meaning. The writing system, **Chữ Quốc Ngữ**, utilizes a dense array of diacritics to represent both vowel quality and tones. This creates a complex orthographic space, requiring high sensitivity from models to distinguish words that differ only by a single diacritic mark, making text normalization essential.

- **Topic-Prominent Syntax:** While following a basic Subject-Verb-Object (SVO) order, Vietnamese is notably topic-prominent, often structuring sentences to emphasize the topic over grammatical subjects. Grammatical relationships are realized through function words (particles like đã, sẽ) rather than inflection. Furthermore, the extensive system of noun classifiers adds a layer of semantic categorization that models must learn to encode effectively.

3.2. Statistical Characteristics of AI-Generated Text

In contrast to human writing, which can be derived from complex reasoning and cultural background, text generated by LLMs is governed by probabilistic sampling from a learned distribution. Statistical inconsistencies that form the basis of detection have been identified in earlier studies:

- **Perplexity and Burstiness:** Human writing shows high variance in the sentence structure and word usage (high burstiness). By contrast, LLMs are designed to optimize the probability of the next token under training data, which tends to steer text towards being statistically ”average”, smoothing patterns out but also making them more repetitive (low perplexity).

- **Syntactic Fluency and Natural Imperfections:** Although modern LLMs generate syntactically fluent text that is often indistinguishable from human-level performance, particularly in languages with large but less-resourced ones such as Vietnamese, but too “perfect” (devoid of the natural typos and slang found in human Q&A forums).

4. Dataset

Currently, resources for detecting AI-generated text in Vietnamese remain scarce. While a few datasets like ViDetect exist, they are often constrained by accessibility issues or opaque generation pipelines, hindering reproducibility. Consequently, we introduce a novel dataset designed to overcome these limitations. The methodology is organized as follows: Section 4.1 outlines the data collection and annotation framework; Section 4.2 elaborates on the prompt engineering techniques employed; and Section 4.4 presents an exploratory data analysis (EDA) to demonstrate the dataset’s diversity and quality.

4.1. Data Collection

To construct a high-quality dataset for AI text detection, we established a rigorous collection pipeline focusing on authentic human-authored content. The data was sourced

from two primary platforms: **Noron.vn** (a prominent Vietnamese Q&A social network similar to Quora or Reddit) and **Wikipedia**. These sources were selected to represent a spectrum of linguistic styles, ranging from the informal, conversational tone of social media to the formal, academic style of encyclopedic entries.

To guarantee the ”human” authenticity of the data and eliminate the risk of accidental AI contamination, we imposed a strict temporal constraint: only texts published **prior to 2021** were collected, predating the widespread public adoption of Large Language Models (LLMs) like ChatGPT.

Initially, we crawled 49,136 raw data points (46,842 from Noron.vn and 2,321 from Wikipedia). Subsequently, a manual curation process was applied to filter out low-quality content. We removed texts containing offensive language or incoherent structures. The final human dataset comprises **4,745 samples** (4,265 from Noron.vn and 480 from Wikipedia), serving as the ground truth (Label 0) for our experiment.

4.1.1 AI Data Generation Strategy

To create a balanced dataset, we generated a corresponding set of AI-authored texts (Label 1) using state-of-the-art LLMs (ChatGPT/Gemini/Gemma-3). To ensure the detection model is robust against different types of AI usage, we categorized the generation process into two distinct strategies [13]:

- **Task 1: Content Generation (Question/Answer Generation):** We prompted the LLM to generate entirely new content based on topics extracted from the human dataset. This simulates scenarios where users employ AI to write essays or answer questions from scratch.
- **Task 2: Content Polishing (Rewriting):** We fed the human-written texts into the LLM and instructed it to ”polish” or ”rewrite” the writing style while retaining the original meaning. This simulates the use of AI as an editing tool, which produces text that is harder to detect due to the preservation of human nuances.

Finally, the combined dataset (Human and AI) was partitioned into training, validation, and testing sets with a ratio of **8:1:1**, respectively. This split ensures sufficient data for model optimization while retaining a separate set for unbiased evaluation.

4.2. Prompt Engineering

To ensure the AI-generated text closely mimics the linguistic patterns of the human reference data, we designed specific prompt templates for each domain. Inspired by methodologies employed in English datasets for Reddit and Wikipedia detection, we adapted these techniques to the Vietnamese context, specifically targeting the distinct stylistic differences between academic writing (Wikipedia) and social media discussions (Noron.vn).

Our prompting strategy addresses two distinct generation tasks:

- **Generation Task (New Content):** Generating entirely new text from a seed topic or question.
- **Polishing Task (Rewriting):** Paraphrasing existing human text to simulate the use of AI as an editing assistant.

4.2.1 Prompt Design Rationale

The prompt complexity required varies significantly by domain. For **Wikipedia**, the prompts are designed to be concise and direct (Zero-shot). Since encyclopedic text follows a standardized, formal structure, simple instructions are sufficient to trigger the model’s academic writing style.

Conversely, for **Noron.vn**, we utilized more elaborate **Role-playing Prompts**. Social media texts often contain slang, personal opinions, specific community jargon, and variable sentence structures. Therefore, the prompts for Noron are significantly longer and more detailed, explicitly instructing the model to adopt the persona of a forum user to capture the informal and conversational nuances of the platform.

4.3. Template Specifications

Inspired by [14], below are designed task prompts for AI data generation and polishing texts on Noron and Wiki.

Sample Prompt for AI Data Generation on Noron

Vai trò: Bạn là một người sáng tạo nội dung trên Noron.
Nhiệm vụ: Đưa ra một câu trả lời sâu sắc và thấu đáo cho câu hỏi dưới đây.
Hướng dẫn:
1. Đưa ra lời giải thích rõ ràng, toàn diện và đi thẳng vào trọng tâm vấn đề.
2. Sử dụng ngôn ngữ đơn giản, đời thường để kết nối với nhiều đối tượng người đọc.
3. Làm nổi bật các ý chính bằng ví dụ cụ thể hoặc câu chuyện thực tế (nếu phù hợp).
4. Giữ giọng văn thân thiện, tự nhiên như một cuộc trò chuyện.
5. Đảm bảo câu trả lời có cấu trúc rõ ràng: mở đầu, thân bài, kết luận.
6. Nếu phù hợp, hãy đưa ra những góc nhìn hoặc quan điểm riêng.

Câu hỏi: "{cau_hoi}"

Sample Prompt for AI Data Generation on Wiki

Hãy viết đoạn mở đầu cho một bài viết Wikipedia có tiêu đề: {tieu_de}.
Chỉ phản hồi bằng câu trả lời được tạo ra, không thêm bất kỳ nội dung nào khác.

Sample Prompt for AI Data Polishing on Noron

Vai trò: Bạn là một người sáng tạo nội dung trên Noron.
Nhiệm vụ: Nhiệm vụ của bạn là biên tập và trau chuốt lại nội dung dưới đây và tuân thủ các hướng dẫn sau:
Hướng dẫn:
1. Đảm bảo nội dung trôi chảy, tự nhiên và thú vị khi đọc.
2. Sử dụng ngôn ngữ đơn giản, đời thường để kết nối với nhiều đối tượng người đọc.
3. Làm nổi bật các ý chính một cách súc tích và ấn tượng.
4. Điều chỉnh giọng văn trở nên thân thiện, mang tính chất trò chuyện cởi mở hơn.
5. Chỉ trả về kết quả là nội dung đã chỉnh sửa, không kèm theo bất kỳ phản hồi nào khác.

Nội dung: "{noi_dung}"

Sample Prompt for AI Data Polishing on Wiki

Hãy viết lại đoạn mở đầu của bài viết Wikipedia đã cho theo một cách diễn đạt khác.
Chỉ phản hồi bằng câu trả lời được tạo ra, không chào hỏi hay nhận xét.
Dưới đây là đoạn mở đầu gốc: "{noi_dung}"

4.4. Data Statistics

After data collection and preprocessing, we have obtained a complete dataset to serve the AI-generated text detection problem. Based on the details of the dataset presented in Table 1, a primary observation is the significant verbosity of AI-generated content, which possesses the highest average length (2,295.33 characters)—nearly double that of human writing (1,315.87 characters). This supports the hypothesis that LLMs tend to be more elaborate and comprehensive when generating new content.

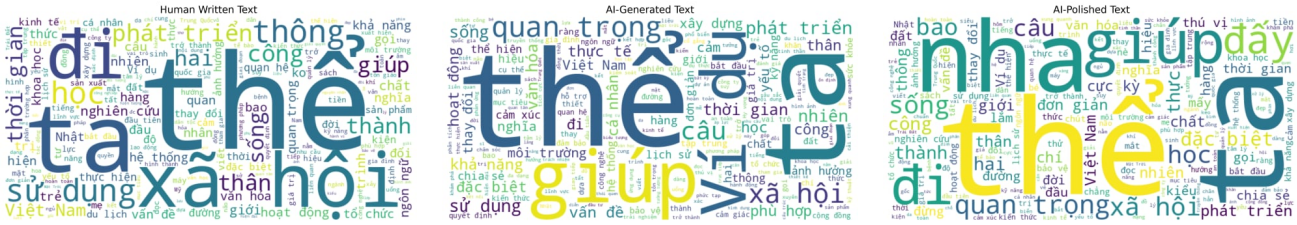
In contrast, the AI-polished texts exhibit the shortest average length (1,212.14 characters), suggesting that the rewriting process tends to streamline content and remove redundancies found in the original human drafts. Furthermore, Human Writing demonstrates the highest variability, evidenced by the largest standard deviation (1,428.29) and an extreme maximum length of 26,515 characters. This reflects the diverse and unpredictable nature of human expression, whereas AI outputs generally adhere to more consistent length patterns.

Table 1: Detailed statistics of the dataset across three categories.

Metric	Human Writing	AI Generated	AI Polished
Size (samples)	4,745	4,745	4,745
Avg. Length	1,315.87	2,295.33	1,212.14
Min. Length	150	322	151
Max. Length	26,515	8,261	7,782
Std. Dev.	1,428.29	1,171.88	664.92

Lexical Analysis: Figure 1 highlights distinct lexical signatures after removing common Vietnamese stopwords.

- **Human Writing (Fig. 1a):** The vocabulary is deeply grounded in personal narratives and daily activities. Prominent keywords include pronouns like *"mình"*, *"bạn"* and active verbs such as *"đi"*, *"học"*, and *"sống"*. This reflects the subjective and experiential nature of human social media posts.
- **AI Generated (Fig. 1b):** The AI model exhibits a "functional" and "explanatory" tone. The dominant word is *"giúp"*, reflecting the assistant-like persona of LLMs. Other frequent terms include *"phát triển"*, *"quan trọng"*, and *"thông tin"*, indicating a tendency towards generalized and academic expression.



(a) Human Writing **(b)** AI Generated **(c)** AI Polished

Figure 1: Lexical comparison using Word Clouds

- **AI Polished (Fig. 1c):** The polishing task creates a hybrid lexical distribution. While it retains core human topics (e.g., *"sống"*, *"bạn"*), it introduces AI-typical "enhancers" such as *"đặc biệt"* (especially) and *"thay đổi"* to restructure the sentences, effectively bridging the gap between casual human speech and formal machine structure.

5. Methodology

5.1. Data Pre-processing

To optimize the performance and interpretability of our AI text detection model, we implemented a rigorous pre-processing pipeline designed to eliminate noise and mitigate bias. The specific techniques are as follows:

- **Noise Reduction and Normalization**: We systematically removed non-semantic elements, including emojis, markdown syntax (e.g., asterisks, bolding), and excessive whitespace. This step reduces feature sparsity and ensures that the model focuses on linguistic content rather than encoding formatting irregularities which may vary arbitrarily between platforms.

- **Removing stop words**: We removed common words in the Vietnamese language that do not have important meanings.

- **Mitigation of AI Artifacts (De-biasing)**: Large Language Models (LLMs) often exhibit specific "signatures," such as repetitive structural markers (e.g., "Mở đầu," "Kết luận") or standard refusal/conversational phrases (e.g., "Với tư cách là một trợ lí ảo," "Dưới đây là câu trả lời"). To prevent the detector from learning spurious correlations - where the model relies on these trivial keywords rather than analyzing the actual writing style - we employed heuristic filters to strip these explicit markers from the AI-generated samples.

- **Filtering Low-Context Samples**: We excluded samples with a length of fewer than 150 characters. As noted in previous studies [14], extremely short texts lack sufficient semantic density and stylistic signals required for accurate detection. Filtering these samples reduces noise in the training data and ensures the model is trained on inputs with adequate context to discern complex linguistic patterns.

5.2. General Fine-tuning Framework

To ensure suitability for the Vietnamese language and optimal performance on the dataset, we follow the fine-tune models method. We formulate the detection task as a binary classification problem where the input text X is passed through a pre-trained backbone encoder M to obtain a contextual representation vector H. For encoder-only models, we extract the vector corresponding to the special classification token (e.g., `<s>` or `[CLS]`). This vector is then fed into a linear classification head:

$$\hat{y} = \text{Softmax}(W \cdot H_{CLS} + b)$$

For encoder-decoder models, we utilize the encoder's output to predict the label.

5.3. Investigated Pre-trained Models

We investigated diverse state-of-the-art Vietnamese language models and multilingual models to serve as backbones for our detector. The selected models cover different architectures and pre-training data scales as follow:

- **PhoBERT-based**: PhoBERT [15] is a **RoBERTa**-based model trained on 20GB of text. It uses dynamic masking and is optimized for Vietnamese word segmentation, serving as the standard baseline for NLU tasks in Vietnam.

- **CafeBERT**: **CafeBERT** [16] is developed based on the XLM-RoBERTa model, with an additional pre-training step utilizing a signifi cant amount of Vietnamese textual data to en hance its adaptation to the Vietnamese language

- **ViT5-base**: Unlike BERT-based models, ViT5 [17] is a Text-to-Text Transformer developed by VietAI. It follows the **T5** (Text-to-Text Transfer Transformer) architecture, treating every NLP problem as a generation task. Trained on a massive and diverse corpus (CC100), ViT5 excels in capturing the generative patterns of the Vietnamese language.

- **BARTpho-base**: Another contribution from VinAI, BARTpho [18] utilizes the **BART** architecture, which functions as a denoising autoencoder. While it is a sequence-to-sequence model designed primarily for generative tasks, its encoder component provides robust contextual representations, making it a strong candidate for classification when fine-tuned.

- **mDeBERTa-v3-base**: To benchmark against multilingual capabilities, we employ mDeBERTa-v3 [19], a state-of-the-art model by Microsoft. It improves upon the original BERT and RoBERTa by using **disentangled attention** and an **ELECTRA-style** pre-training objective with Gradient Disentangled Embedding Sharing. This allows the model to capture subtle syntactic and semantic nuances across 100 languages, including Vietnamese, with high efficiency.

6. Experiments and Results

6.1. Evaluation Metrics for LLM-generated Text Detection

Comprehensive evaluation requires multiple metrics:

- Accuracy: Overall correctness (limited utility with imbalanced data)
- Precision: $\frac{\text{True AI}}{\text{Predicted AI}}$ – measures false alarm rate
- Recall: $\frac{\text{Detected AI}}{\text{Actual AI}}$ – measures detection completeness
- F1-score: Harmonic mean of precision and recall: $2 \times \frac{Precision \times Recall}{Precision + Recall}$

Proper evaluation requires strict separation of training, validation, and test sets, with the test set containing previously unseen AI models and prompt variations.

6.2. Experimental Configuration

All models were fine-tuned using the Adam optimization. We utilized an NVDIA-A100 setup with 40GB of memory to fine-tune models. The hyperparameters for the models are set as follows: epochs = 10, learning rate = 2e-05, dropout = 0.1, batch size = 32, and early stopping patience = 3, max_len = 256 tokens

6.3. Experimental Results

As presented in Table 2, the investigated models demonstrate exceptional performance on the Vietnamese AI detection task, with an average F1-score of 96.12% on the **combined test set (mixed of AI-Generated and AI-Polished)**

- **Impact of Model Architecture:** Among all candidates, **CafeBERT** achieves state-of-the-art performance, recording the highest scores across all metrics and categories (Combined F1: 98.54%). This superiority can be attributed to its hybrid pre-training strategy, which combines multilingual generalization with extensive Vietnamese-specific fine-tuning. **ViT5** and **BARTpho** also perform competitively (Combined F1 of 97.38% and 97.00%, respectively), demonstrating that sequence-to-sequence architectures are highly effective for this classification task.

- **Impact of Language-Specific Pre-training:** A significant performance gap is observed between Vietnamese-centric models and the multilingual baseline. **mDeBERTa**, despite being a robust multilingual model, lags behind significantly, achieving a Combined F1-score of only 90.74% - approximately 6-8% lower than its Vietnamese counterparts. This finding emphasizes that language-specific pre-training is crucial for capturing the subtle linguistic nuances required for accurate detection in Vietnamese.

- **Difficulty of "Polishing" Task:** Consistent with our dataset analysis, the "AI Polished" category proves to be more challenging than "AI Generated." All models exhibit a noticeable performance drop when detecting polished texts. For instance, **PhoBERT**'s F1-score decreases from 99.04% on the Generated set to 95.30% on the Polished set. This indicates that when AI is used to edit or rewrite human text, it retains a significant portion of the original human stylistic features, making it harder for classifiers to distinguish.

Table 2: Performance comparison of models across three datasets: AI-Generated, AI-Polished, and Combined

Model	AI Generated				AI Polished				Combined			
	Acc	Pre	Rec	F1	Acc	Pre	Rec	F1	Acc	Pre	Rec	F1
PhoBERT	99.04	99.04	99.04	99.04	95.31	95.48	95.31	95.3	97.27	97.17	97.68	96.92
CafeBERT	99.68	99.68	99.68	99.68	97.04	97.05	97.04	97.04	98.7	98.72	98.37	98.54
ViT5	99.57	99.58	99.57	99.57	96.84	96.84	96.84	96.84	97.68	97.63	97.14	97.38
BARTpho	99.47	99.47	99.47	99.47	95.61	95.62	95.61	95.61	97.34	97.03	96.98	97.00
mDeBERTa	93.19	93.20	93.19	93.19	90.00	90.36	90.00	89.98	91.55	89.90	91.92	90.74
Avg.	**98.19**	**98.19**	**98.19**	**98.19**	**94.96**	**95.07**	**94.96**	**94.95**	**96.51**	**96.09**	**96.42**	**96.12**

7. Demo Web Application

To demonstrate the practical applicability of our best-performing model (CafeBERT), we developed a real-time AI text detection system using React.

As illustrated in Figure 2, the interface features a minimalist design that allows users to input Vietnamese text segments. The system processes the input through the fine-tuned CafeBERT model and returns classification results along with prediction probabilities (Human-written or AI-generated/ AI-polished).

The inference latency is optimized, averaging only about 200ms per request in RTX 5070 GPU environment, ensuring a smooth user experience. This web system not only effectively supports users but also aims toward the future integration of text detection capabilities on social media platforms.

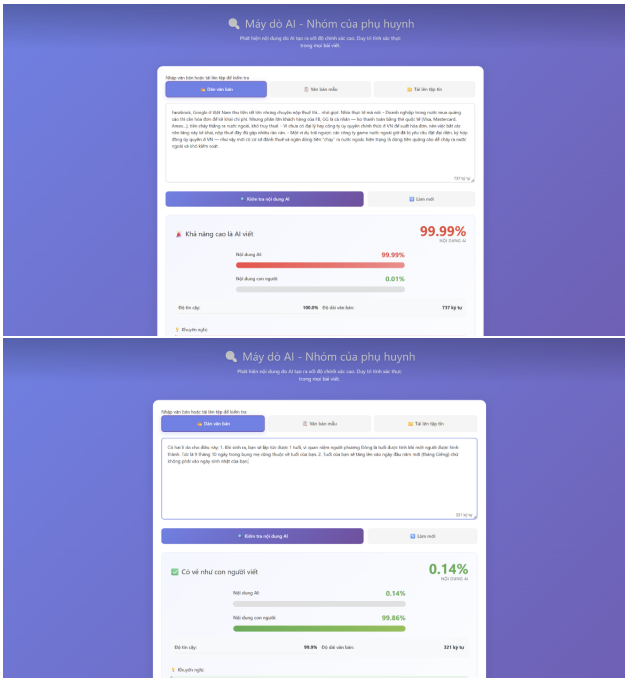


Figure 2: Detection Result

8. Challenges and Limitations

8.1. Challenges

Despite the promising results, our study encounters significant hurdles inherent to the adversarial nature of the field and the specific characteristics of the target language.

The primary challenge stems from **the evolving landscape of Generative AI**, which creates an inevitable "arms race" between generation and detection. With the continuous improvement of LLM, the statistical gap between human-authored and automatically generated text diminishes, rendering detection based solely on linguistic patterns increasingly difficult. This issue is compounded by adversarial strategies, such as paraphrasing, style transfer, and hybrid human-AI co-writing, which intentionally obfuscate the "watermarks" or distinctive fingerprints typically left by generative models.

Furthermore, **linguistic complexity and data scarcity** complicate the detection task beyond algorithmic constraints. Additionally, the detection problem is challenging due to **Vietnamese typological properties**. In social media text, code-switching (the mixture of English and Vietnamese) as well as transferred words prevail, which adds noise into the system and may give rise to a false positive. Besides, the necessity of correct representation of diacritic marks and tonal variations requires high sensitivity models which are circumscribed by limited availability of large-scale, well-annotated annotated data for Vietnamese AI detection as for high-resource languages such English.

8.2. Limitations

While our framework yields robust results, we acknowledge specific limitations regarding **generalizability** and **scalability**. First, the size of the dataset is still limited relative to existing English benchmarks and narrowly restricted to encyclopedic and social media contexts. As a result, how the model performs on unseen genres (e.g., creative literature or law documents) remains unverified - linguistic features might not transfer well to these specialized domains.

Furthermore, our evaluation relies on a fixed subset of LLMs available at the time of data collection. **The system has not yet been stress-tested against emerging proprietary architectures or specific Vietnamese-centric models**, potentially leaving a blind spot regarding the statistical fingerprints of newer generation engines. Finally, the detector acts as a black box binary classifier that lacks interpretability: it does not output span-level explanations of how the input text is attributed to a certain source model, limiting its applicability in scenarios where explainability is a critical requirement

9. Conclusion and Future Work

This report has surveyed the landscape of AI-generated Vietnamese text detection on Social Media. Utilizing a VSM-D comprising over 4,745 data samples from Noron and Wikipedia after generated and polished from the initially data from humans to ensure the results enough trust for real-world application. Moreover, the study can assist and is a base for other study about LLM on social media.

In the future, we will enhance data and analysis about the use of AI trend on Social Media to evaluate the quality of contents from human and assist users write text on social media more emotional and empthy with other people.

Future research should prioritize:

- Hybrid approaches: Combining statistical, ML, and deep learning methods
- Vietnamese-specific features: Leveraging tonal, syntactic, and cultural cues
- Continual learning: Systems that adapt to evolving LLM capabilities
- Explainable detection: Providing interpretable rationale for classification decisions
- Standardized benchmarks: Community-agreed evaluation datasets and protocols

As LLMs become increasingly sophisticated, the development of robust, fair, and effective detection systems for Vietnamese will remain an essential component of responsible AI deployment in Vietnamese-speaking contexts.

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